

CE716 – PROJECT MANAGEMENT AND CONTROLS

Assignment 01

Drawing Review, Scope Definition and Preliminary Project Planning

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1. Basis of Review and Key Assumptions

Drawing basis. The assessment uses the architectural Ground/First/Second Floor Plan and the structural Plinth Beam Plan, Ground Roof Slab and First Roof Slab drawings supplied with the assignment.

Area assumption. The architectural drawing shows an overall plan dimension of approximately 47'-7" × 26'-7". This is treated as the approximate building footprint, not necessarily the legal plot boundary. Hence the exact plot area, setbacks and sanctioned built-up area cannot be confirmed from the supplied drawings alone.

Calculated planning area. 47'-7" × 26'-7" ≈ 1,265 sq ft per full floor plate. Assuming a similar gross footprint for Ground + First + Second floors gives approximately 3,795 sq ft gross floor area. Actual sanctioned/chargeable built-up area may differ because of balconies, parking, projections and local regulations.

2. (a) Scope of Work – What Needs to Be Done

2.1 Project description

- Private residential independent house/villa.
- Ground + 2 floors (G+2). The structural drawings explicitly state that the design is for G+2 only.
- Approximate structural footprint: 47'-7" × 26'-7" ≈ 1,265 sq ft.
- Approximate gross floor area for three levels: ≈ 3,795 sq ft, subject to verification against the sanctioned plan.
- Approximately 5 bedrooms, living room, dining area, kitchen, multiple toilets/dressing areas, stores, balconies, parking and a dedicated home-theatre room on the second floor.
- RCC structural system with beams, columns and slabs; the structural drawings show different beam/slab types, sunk slab zones and special structural members.

2.2 Major activities / work packages

Work Package	Major Activities
Pre-construction	Verify title/ownership, site survey, benchmarks, plot dimensions, levels, setbacks, approvals and NOCs; appoint consultants and contractor.
Design coordination	Coordinate architecture, structure, plumbing, electrical, HVAC, rainwater, septic/soak-pit and home-theatre requirements; freeze drawings and specifications.
Tendering & procurement	Prepare scope/BOQ, prequalify contractors, obtain comparable quotations, negotiate and finalize contract, programme and payment terms.
Site establishment	Fencing, temporary water/electricity, storage, safety facilities, setting-out and construction benchmarks.

Earthwork & foundation	Excavation, PCC where specified, footing/foundation reinforcement and concrete, column starters, backfilling and compaction.
Plinth	Plinth beams, filling, damp-proofing/waterproofing, plinth slab and associated services.
RCC frame	Columns, beams and slabs for G+2, including ground roof and first roof slab systems, formwork, reinforcement, concreting, curing and de-shuttering.
Masonry & architectural shell	Walls, partitions, lintels/sills, staircase, door/window openings, parapets and associated civil works.
MEP	Electrical, plumbing/drainage, sanitary fixtures, water supply, septic/soak pit if applicable, rainwater, pumps, HVAC and home-theatre services.
Finishes	Plastering, flooring/tiles, waterproofing, painting, doors, windows, railings, sanitary/electrical fixtures and kitchen works.
External works	Parking/paving, boundary wall/gate, drainage, rainwater management and landscaping as required.
Testing & handover	System testing, waterproofing checks, snag rectification, as-built documents, warranties and completion/occupancy documentation.

3. (b) Information Missing Immediately Before Starting Work

The architect's statement that the drawings are sufficient to start work should not be accepted without a multidisciplinary completeness check. The following information is required before committing the owners to construction:

Missing Item	Why Needed Immediately
Sanctioned architectural package	Approved/sanctioned plan, site plan, setbacks, floor levels, elevations, sections, staircase details, door/window schedule and opening sizes.
Site survey	Authenticated plot boundary, dimensions, existing levels, road level, neighbouring structures, access, utilities and permanent benchmark.
Geotechnical/soil report	Bore logs, groundwater and design SBC. The supplied structural sheets conflict: the Plinth Beam Plan states 20 T/m ² at 3 m, while the Ground Roof and First Roof Slab sheets state 15 T/m ² at 3 m. Structural engineer must resolve this.
Complete foundation design	Footing layout, sizes/depths, reinforcement, pedestals/column starters, founding level and foundation sections.
Complete structural package	Column schedule/reinforcement, all beam/slab schedules, staircase design, lintel/chajja details, structural sections and design basis/calculations.
MEP drawings	Electrical load/layout, plumbing/drainage, rainwater, septic/soak pit, water supply, pumps/tanks, HVAC provisions and home-theatre services.
Construction specifications	Concrete/steel/masonry/finish specifications, approved-equivalent criteria, waterproofing specification and testing requirements.

BOQ/tender information	Measured quantities or sufficiently clear scope, item descriptions, exclusions, taxes and payment milestones.
Legal/approval status	Building permit/sanction, applicable NOCs, work-start requirements and confirmation that the latest approved revision is being issued for construction.
Contractor scope/programme	Contract conditions, construction schedule, manpower/equipment plan, safety plan, quality plan and responsibility matrix.

4. (c) Other Missing Information and When It Will Be Needed

Information	When Needed	Reason
Final finishes & interior selections	Before tendering/finishing procurement.	Tiles, stone, sanitaryware, doors/windows, kitchen, paint, lighting and premium fixtures affect cost and programme.
Electrical load & equipment schedule	Before electrical design freeze and procurement.	ACs, geysers, pumps, kitchen equipment and home theatre determine electrical loads and points.
HVAC design	Before slab/false-ceiling and electrical coordination.	Sleeves, drains, outdoor-unit locations and power requirements must be coordinated early.
Waterproofing details	Before toilets, balconies, sunk slabs and roof works.	System, slopes, outlets, testing and warranty requirements must be fixed.
Home-theatre design	Before second-floor finishes and rough-ins.	Screen, projector, acoustics, speakers, ventilation, lighting and AV wiring need coordination.
Kitchen design	Before plumbing/electrical rough-in.	Counter, sink, appliances, sockets, exhaust/chimney and service points depend on final layout.
External works/landscape	Before final external development.	Levels, drainage, parking paving, rainwater, gate and boundary works must be coordinated.
Long-lead procurement plan	Before each procurement package.	Doors/windows, specialty sanitaryware, electrical panels, theatre equipment and custom finishes can affect schedule.
As-built documentation	Throughout construction and before handover.	Record final dimensions, concealed services, changes, warranties and equipment information.

5. (d) First Agencies / Contractors to Engage

- 1. Licensed land surveyor:** Confirm plot boundaries, dimensions, levels, north and setting-out control points.
- 2. Geotechnical/soil-testing agency:** Confirm soil profile, groundwater and foundation parameters; reconcile the SBC discrepancy.

3. Structural consultant/engineer: Complete/review foundation and RCC package and resolve discrepancies before RCC work.

4. Architect + MEP consultants: Coordinate architectural, structural, plumbing, electrical, HVAC, rainwater and special-room requirements.

5. Main civil/RCC contractor: Prequalify at least three competent contractors, obtain comparable bids and select on capability, price, resources, safety and past performance.

6. Specialist subcontractors/vendors: Electrical, plumbing, waterproofing, aluminium/uPVC, HVAC and home-theatre specialists after designs are sufficiently frozen.

6. (e) Preliminary Construction Cost – Ballpark Estimate

Method. No detailed quantity take-off is attempted because the drawing package is incomplete. For planning, approximately 3,795 sq ft gross floor area is used.

2026 market context. Current published Kanpur references place standard construction around ₹1,450–₹1,850/sq ft and premium construction around ₹1,900–₹2,500/sq ft. Another 2026 benchmark gives a standard range of about ₹1,710–₹2,215/sq ft. These are indicative market references, not tender rates. Because this house has G+2 construction, multiple bathrooms, sunk slabs, a home theatre and special structural elements, a planning allowance above basic construction is appropriate.

Scenario	Rate	Approx. Cost	Comment
Standard benchmark	₹1,450–₹1,850/sq ft	₹55–70 lakh	Lower-bound reference; may understate premium finishes.
Premium benchmark	₹1,900–₹2,500/sq ft	₹72–95 lakh	Reasonable good-quality RCC residential benchmark.
Recommended planning allowance	₹2,300–₹2,800/sq ft	₹87–106 lakh	Working allowance for this project's special features; this is an inference, not a quoted rate.
With ~10% contingency	Approx. ₹2,530–₹3,080/sq ft equivalent	₹96–117 lakh	Budget envelope until BOQ and quotations are available.

Recommended answer to Chandler: Tentatively budget about ₹1.0–₹1.2 crore for construction, excluding land, financing costs and unusually high-end loose furniture/interiors. This is a planning range, not a commitment; it should be replaced by a BOQ-based estimate after design freeze and contractor quotations.

7. (f) Proposed PMC Fee and Why the Family Should Hire Us

Proposed fee: 3% of the final construction cost, subject to a minimum fee of ₹3 lakh, with reimbursable expenses defined separately. For a ₹1.0–₹1.2 crore construction budget, this is approximately ₹3.0–₹3.6 lakh.

Market check: Published 2026 Indian PMC references commonly place residential PMC fees in the low-single-digit range; examples give 1–3% and 1–3.5%. The proposed 3% is therefore defensible for a full-scope PMC on a relatively complex G+2 residence, subject to a written scope.

7.1 Value proposition

- Joey: the PMC is not another contractor. It represents the owners, checks measurements and quality, and helps prevent rework, poor workmanship, unapproved substitutions and avoidable overruns.
- Chandler: we establish and track the budget, compare contractor bids, verify measurements/RA bills, control variations and provide early warning on cost and schedule.
- Monica: drawing revision control, material approvals, inspection checklists, mock-ups and snag lists protect the details she cares about.
- Rachel: premium finishes and specialist packages can be planned transparently, separating essential construction from optional luxury upgrades.
- Ross: the library requirement should be frozen and coordinated for structural loading, shelving, lighting and electrical points.
- Phoebe: a requirements/decision register will convert unclear requirements into explicit decisions, assumptions or deferred items.

7.2 Why PMC instead of self-management?

- Construction requires continuous coordination among architect, structural engineer, MEP consultants, contractor, suppliers and specialist vendors.
- Defects in RCC, waterproofing and concealed services are costly to discover after work is covered.
- Competitive tendering and measurement-based payments reduce the risk of an apparently low quote becoming expensive through variations.
- Programme tracking gives the owners early warning about procurement, labour or design decisions that threaten completion.
- The PMC provides one owner-side coordination point and maintains records for future maintenance and handover.

8. (g) Other Comments / Concerns for the Next Meeting

- Freeze requirements in a written room-by-room matrix. Phoebe's unclear requirements must be converted into decisions before design changes accumulate.
- Resolve the conflicting SBC information: 20 T/m² in the Plinth Beam Plan versus 15 T/m² in the Ground Roof and First Roof Slab drawings.
- Confirm whether 47'-7" × 26'-7" is the legal plot boundary or only the building footprint; this affects setbacks, parking, approvals and area calculations.
- Recheck the ground-floor parking arrangement against the intended vehicle because the architectural drawing shows a compact parking area.
- Confirm that architectural, structural and MEP drawings are coordinated and carry the same revision.
- Confirm whether each drawing is Draft, GFC or As-Built. Construction should use the latest approved/GFC issue.
- Confirm staircase dimensions, headroom, handrails and structural details before execution.
- Confirm all toilet/sunk-slab locations and plumbing routes before casting relevant slabs.
- Freeze home-theatre requirements early because acoustics, HVAC, electrical and AV conduits affect the building fabric.
- Agree on budget priorities so Chandler's cost control and Rachel's luxury expectations can be managed through a transparent base-plus-options approach.

- Implement written change control: no site change without documented scope, cost/time impact and owner approval.
- Agree on duration, payment milestones, retention, defect-liability obligations, insurance, safety responsibilities and dispute resolution.

9. Recommended Immediate Action Plan

- Day 1–2: Create a drawing register and mark every missing/unclear document.
- Day 1–3: Surveyor and soil-testing agency verify site conditions and foundation parameters.
- Day 2–5: Structural engineer reconciles foundation/SBC information, columns, staircase and special structural elements.
- Day 3–7: Architect/MEP team completes missing coordination drawings and specifications.
- Day 5–8: Prepare preliminary BOQ/scope and contractor tender package.
- Day 7–12: Prequalify and obtain comparable bids from at least three main contractors.
- Day 10–14: Technical and commercial bid evaluation; negotiate and finalize contractor.
- After approvals/design freeze: Mobilize the site and commence construction.

10. Conclusion

The drawings are sufficient to understand the broad concept of a G+2 residential house, but they are not sufficient by themselves to responsibly mobilize construction. The major missing inputs are the sanctioned architectural package, complete foundation/column structural information, soil report, coordinated MEP drawings, specifications, BOQ/tender documents and approval status. The immediate PMC role should therefore be to close these information gaps, freeze scope, competitively procure the contractor and establish cost, quality and time controls.

Planning recommendation. Use approximately 3,795 sq ft as the preliminary gross floor area and a provisional construction budget of about ₹1.0–₹1.2 crore, excluding land and unusually high-end loose interiors. Convert this to a measured BOQ and contractor-backed estimate before award.

11. References Used for Current-Market Context

- Kanpur 2026 construction-cost benchmark: Ghar Ka Budget, July 2026 – standard ₹1,450–₹1,850/sq ft; premium ₹1,900–₹2,500/sq ft.
- Kanpur 2026 construction-cost benchmark: InfraLens, July 2026 – standard approximately ₹1,710–₹2,215/sq ft.
- 2026 PMC fee references: InfraLens and Metamorph – residential PMC references in the low-single-digit percentage range.
- UP/Kanpur approval context: Kanpur Development Authority and the Uttar Pradesh Online Building Plan Approval System; current approval route/status should be verified before construction.